

TUNABLE NON-LINEAR MANIFOLD ROMs FOR ELLIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

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ABSTRACT

A reduced-order PDE solver should allow a practitioner to choose an accuracy and computational cost after training. We present a nonlinear manifold reduced-order model (NM-ROM) that provides this choice through nested correction directions added to a frozen decoder. Correction rank changes the trial space without retraining; solver tolerance and, where applicable, empirical quadrature control the cost of the reduced solve. The architecture combines a learned spatial bank with a small nonlinear coefficient map, exact boundary enforcement, and matrix-free weak residual projection. Linear operators are precomputed, while validated quadrature reduces the cost of nonlinear residual evaluation. Experiments on elliptic, parabolic and hyperbolic problems compare the resulting accuracy–cost tradeoff with named full-order solvers; the main comparisons report error and cost against each named FOM. On Burgers, a fixed-test-space correction ladder spans $2.44\times$ in same-grid error for $5.16\times$ in runtime; a separate scheduled ladder is tested across training seeds and a held-out cohort. Acceleration depends on the comparator and resolution: selected reduced solves are faster than iterative full-order methods, while the efficient FOM remains stronger in several cases. These results establish deployment-time tunability and delimit its practical benefit, rather than a universal speed advantage.

1 INTRODUCTION

Neural operators for partial differential equations (PDEs) (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023; Li et al., 2024; Boullé and Townsend, 2024) provide fast amortised inference. At a fixed grid and inference procedure, a trained operator returns one accuracy and runtime point. A practitioner may instead need to spend more computation on a difficult query, or accept a less accurate solution when response time matters. Classical full-order methods (FOMs) expose numerical controls such as iteration tolerance, but their cost can grow substantially with mesh resolution. The question is whether a trained reduced model can provide a useful accuracy–cost tradeoff at deployment.

Projection-based reduced-order models address this question by solving the governing equations in a smaller space (Benner et al., 2015; Sirovich, 1987). A linear subspace is inexpensive to use, but can require many basis functions to represent moving fronts. Nonlinear-manifold ROMs learn a more compact representation (Lee and Carlberg, 2020; Kim et al., 2022); their online nonlinear solve can, however, offset the benefit of fewer unknowns. Representation accuracy and computational advantage therefore need to be measured separately.

We present an NM-ROM with a deployment-time accuracy–cost family from one trained decoder. The solution is represented by a learned spatial bank multiplied by a nonlinear map of latent variables. We augment this map with nested, precomputed correction directions. Activating more directions enlarges the trial space without changing the learned weights. The correction rank is the primary representation control; iteration limits and stopping tolerances govern the numerical solve, and empirical quadrature reduces residual-evaluation cost where its validation permits. All online coefficients are determined from the weak PDE residual, without access to the reference solution.

Our experiments ask whether this mechanism improves accuracy, what it costs, and when it is competitive with established solvers. The comparisons in this paper use full-order solvers. Full-order

timings name both the algorithm and its settings: each linear-PDE speedup is measured against the displayed CG setting. The main section includes two- and three-dimensional experiments, with held-out final results distinguished from development comparisons.

Contributions.

1. **A tunable NM-ROM from one trained model.** Nested correction directions provide discrete deployment settings that change approximation capacity without retraining. We isolate correction rank at fixed residual test count and separately evaluate a prescribed test-count schedule across seeds and held-out cases.
2. **An architecture supporting the accuracy–cost tradeoff.** A learned spatial bank, a nonlinear head with a linear skip, and nested corrections support compact latent solves, boundary enforcement and node-local evaluation. The nonlinear coordinates and linear corrections have distinct roles in the reduced representation.
3. **A practical weak-form implementation and paired evaluation.** Precomputed linear operators, matrix-free differentiation and validated empirical quadrature make the online solve measurable against full-order solvers. Paired accuracy and runtime measurements identify the PDEs, settings and resolutions where acceleration occurs.

2 RELATED WORK

Projection-based and nonlinear-manifold ROMs. Linear ROMs approximate solutions in a fixed subspace (Benner et al., 2015; Sirovich, 1987); moving fronts can require a large basis (Cohen and DeVore, 2015; Greif and Urban, 2019). Nonlinear-manifold ROMs combine learned representations with residual projection (Lee and Carlberg, 2020; Kim et al., 2022), including quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), neural-field representations (Kim et al., 2024) and adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015). Our focus is a nested set of correction directions within one learned bank, selected after training. Least-squares Petrov–Galerkin projection (Carlberg et al., 2011) and preassembly of reduced operators (Ștefănescu and Sandu, 2014; Weder et al., 2024) are established components; the experiment isolates the effect of correction rank.

Neural operators and hyper-reduction. FNO (Li et al., 2021), DeepONet (Lu et al., 2021), U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and Transolver (Wu et al., 2024) learn solution maps. Their evaluation resolution or inference procedure can also change accuracy and cost; our mechanism instead changes the trial space used to solve the PDE. Empirical quadrature (Hernández et al., 2017; Yano and Patera, 2019) reduces nonlinear residual evaluation through non-negative weights. We retain that construction, testing its error on states reached by the solver and through independent re-draws. The main iterative linear-system comparator is conjugate gradient (Hestenes and Stiefel, 1952).

3 METHODOLOGY

We turn a learned manifold into a practical PDE solver through three components: a trial manifold whose decoder enforces Dirichlet conditions exactly and carries nested linear correction directions (§3.1); a least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests (§3.2); and exact preassembly of every linear term with Empirical Quadrature for Burgers advection where validated (§3.3); §3.4 gives the architecture. Throughout, $u \in \mathbb{R}^n$ is the full-order state on a uniform grid of N intervals per axis, $A \in \mathbb{R}^{n \times n}$ the discrete negative Laplacian; R is the bank width, k the latent dimension, M the number of weak tests and m the number of quadrature nodes. The formulas below illustrate scalar Dirichlet problems; periodic vector fields and time-integrator details are specified per PDE in the appendix. Figure B.1 (Appendix B) shows the data flow; Appendix A gives the per-PDE derivations and exit codes.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

We train a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^n$, $k \ll n$, with latent state z and no encoder in the deployed path, separable into a frozen spatial *bank* $G \in \mathbb{R}^{n \times R}$ times a small neural *head* $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$,

$k \leq R \ll n$, the bank built once per mesh from a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020),

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

To strictly enforce the homogeneous Dirichlet condition on the boundary Γ , we utilize the smooth vanishing factor μ , zero on Γ , folded into every column of the bank, so that $\partial u / \partial z = 0$ on Γ at every resolution (homogeneous data only).

Nested correction directions. The deployed model augments the head’s output with q fixed directions $C_q \in \mathbb{R}^{R \times q}$ and solves for the latent code and the correction coefficients together,

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

The columns of C_q are chosen offline from training data; the Burgers construction uses principal components of the head’s coefficient errors. Per-PDE constructions are stated in Appendix A. The directions are ordered so that C_q is a prefix of $C_{q'}$ for $q < q'$; the ladder is nested and no rung needs retraining. At $q = 0$ the model is the frozen head; at $q = R$ the head is irrelevant and the representation is the full linear span of the bank. For linear PDEs its coefficients can be found by a reduced linear solve; for nonlinear PDEs the equations can remain nonlinear even in this full-bank representation; in between, q moves the reachable set from the head’s image towards the bank’s span, and the solved dimension from k to $k + q$. Two consequences follow: the bank’s projection error is a floor on the achievable error whatever the head or the solver does, so nonlinearity in h_θ buys a smaller *solved* dimension, not an escape from the span; and all x -dependence factors through G , which makes node sampling and exact preassembly available from one decoder.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

Each PDE we consider gives a discretised residual $r(u)$ that should vanish at the true solution. We substitute the trial manifold $u(z, y)$ into r and project onto a fixed, latent-independent test space. On the Dirichlet square, $P \in \mathbb{R}^{M \times n}$ collects low-frequency tensor-product sine vectors, eigenvectors of the five-point operator, $PA = \Lambda P$. The reduced problem solved online is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

with a diagonal row scaling $\Lambda_\star$ stated per PDE in Appendix A: a least-squares Petrov–Galerkin condition with an explicit test space (Carlberg et al., 2011; Lee and Carlberg, 2020), not tangent Galerkin, overdetermined rather than square, solved by a method appropriate to each PDE’s structure.

Elliptic (Poisson). The Poisson residual is $r(u) = Au - f$, and projection gives $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, $B_0 = PG$ (Appendix A.1); the projector onto $\text{range}(B_0 C_q)$ does not depend on z , so y is eliminated exactly (Golub and Pereyra, 1973), the iteration stays k -dimensional at every q , and $q = R$ is a linear solve.

Parabolic (heat). The heat semi-discretisation $du/dt = -\kappa Au$ is advanced with Crank–Nicolson; the reduced step substitutes the manifold into the fully discrete equation *before* projecting and solves $z_{n+1} = \arg \min_z \|B_0 h_\theta(z) - D B_0 h_\theta(z_n)\|_2$ (Appendix A.2), a nonlinear least-squares problem, not a linear system; at $q = R$ the trajectory is the exact modal propagation of the bank coefficients.

Hyperbolic (Burgers). For $u_t + u(u_x + u_y) = \nu \Delta u$ with a sign-upwind stencil and backward Euler (Appendix A.3) the weak residual is nonlinear in the coefficients, so y is not eliminated in closed form; we damp the (z, y) blocks separately inside one Levenberg–Marquardt step (*block-damped* variable projection), which, in the recorded solver comparison, removes the joint LM configuration’s 6 budget exits (Table D.3).

Solver and exits. Each attempt solves the damped normal system $(H + \lambda \text{diag}(\text{diag } H)) \delta = -g$, $H = J^\top J$, $g = J^\top r_w$, accepting the step only if it strictly decreases the residual: damped Levenberg–Marquardt with a monotone test (Marquardt, 1963). Three exit families are recorded and never conflated: the scale-free stationarity measure $\eta(z, y) = \|J^\top r_w\|_2 / (\|J\|_F \|r_w\|_2) \leq \eta_{\text{tol}}$ (4), a residual threshold, and the stalls, reported as early-stopped, never as converged. No query uses the solution it predicts: the elliptic solve starts from the cached training code nearest the projected source, the Burgers query fits (z, y) to the supplied initial field (Appendix A.5).

3.3 HYPER-REDUCTION

The non-linear manifold reduces the DOF count from n to $k + q$, but a projected term such as $PN(Gc)$ still costs $O(n)$. For a fixed linear operator the tested residual is exactly precomputable, $P(Au - f) = B(h_\theta(z) + C_q y) - b$ with $B = \Lambda PG$ assembled offline by the corresponding discrete transforms, so linear terms are never approximated; the Burgers advection $PN(Gc)$ is the only term that resists preassembly, evaluated *densely*, $O(nR)$ per residual, or on an *empirical quadrature* rule of m nodes (Hernández et al., 2017; Yano and Patera, 2019), $O(mR)$. The rule is a non-negative *per-node* weight vector, *independent of the snapshot*, fitted by NNLS (Lawson and Hanson, 1974) to reproduce the projected advection term at n_{fit} stored codes with a hard cap of m nodes (Appendix A.4); changing m re-solves the fit, rules are not nested, and deployment selects among stored rules. *A rule is never accepted on its NNLS fit residual*; we score it by the held-out relative error ρ of the projected advection term (10) over states the solver actually reaches on trajectories disjoint from the fit and evaluation cases, against a primary bar $\rho_{\text{max}} \leq 0.116$ fixed before any certification job ran and a tight bar 0.06, and then re-draw the construction: a rule is *confirmed* only if every re-draw passes (§6.3).

3.4 MODEL ARCHITECTURE

Three properties of the problem motivate our architecture: the latent iteration is initialised cold, so the decoder Jacobian must retain a well-conditioned linear component, which motivates a *linear skip* in the head; hyper-reduction retains a sparse subset of mesh nodes, so the decoder must evaluate at any node in mesh-independent time, which motivates a *per-node bank*; and the family must reach from the head’s image to the bank’s whole span without retraining, which motivates the nested *correction directions*.

Linear skip, for cold-start convergence. The head is a two-layer SiLU MLP plus a linear skip, $h_\theta(z) = \varphi_\theta(z) + W^\top z$, so its Jacobian retains a latent-independent component along which the cold solve can descend; the elliptic solver checks the numerical rank of the projected Jacobian at every query. The skip is a design choice, not ablated here.

No encoder; per-node bank. No encoder is needed: the query fits (z, y) against the supplied input. The decoder restricted to a rule’s support is a cached block times the coefficients, and the network’s parameters do not depend on the mesh.

What is fixed, what is chosen, and how error is reported. Every operating point uses one frozen artefact per PDE; selected at run time are the rank q , the quadrature (dense or a stored rule) and the stopping tolerance and budget; M is held fixed along the headline ladder. The representation ablations distinguish three quantities: the *bank floor* (projection error onto range G), the *best-found* error (the smallest any point of the augmented manifold attains, from a multistart oracle) and the *solved* error the iteration returns.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

Every linear term is preassembled once per mesh as the $M \times R$ matrix B , so the online residual and Jacobian of a linear PDE are $B(h_\theta(z) + C_q y) - b$ and $B[Dh_\theta \ C_q]$, with Dh_θ from a forward-mode Jacobian-Vector Product (Bradbury et al., 2018); the sampled Burgers advection uses the cached stencil block. The damped normal system is solved directly; no Krylov solve is applied to the projected operator.

4.2 TRAINING PROTOCOL

The bank and the head are trained in two stages, both as auto-decoders: the coordinate network g_ϕ is fitted to the training states through (1), then frozen, and the head is fitted with a code library Z on the bank-projected states; C_q is then constructed from training-only coefficient directions as specified for

each PDE. Sizes and cohorts are in Appendix C; checkpoint hashes and recorded offline costs are retained with the source evidence described there.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

Problems and references. The completed two-dimensional study includes viscous Burgers, Poisson, heat, reflective waves and incompressible Navier–Stokes. Poisson is also tested on an L-shaped domain using a CG full-order comparator. Table C.1 gives meshes and cohorts; Table C.2 specifies the sampled families. Burgers provides the principal correction-rank study. Poisson, heat and waves test the benefit of nonlinear restriction for linear equations. Navier–Stokes and lower-viscosity Burgers test its limitations. The three-dimensional comparisons are included in the main results; evaluation scope and reproducibility details are retained in the appendix.

Accuracy. We report worst relative L^2 error over the stated cases and requested times, in percent. Poisson uses the steady reference norm; heat and wave displacement use the current reference norm. Burgers uses the initial-field norm and distinguishes all-times error, including input compression, from evolved-times error. Wave displacement alone does not establish velocity or energy accuracy. Same-grid error measures reduction error relative to the converged discrete solution. Error against a refined reference also includes discretisation error; the two are never interchanged. Development results and held-out tests are labelled separately. Settings for a held-out test are fixed before its cases are accessed.

Timing and speedup. Every accuracy–runtime pair comes from the same solver invocation. We use double precision, GPU burn-in, synchronised timing and medians of retained repetitions. GPU query time includes initialization, the solve or rollout, and requested device outputs. Complete-query time additionally includes host transfers and is labelled separately. Training and compilation are offline costs. Speedup is $S = T_{\text{FOM}}/T_{\text{method}}$, using measurements from the same allocation. Each table identifies the FOM algorithm, tolerance and error. Where a table selects the fastest tested passing FOM with no greater error than a method, that selection and the possibly differing denominators are explicit. A speedup is not an accuracy-equivalence claim. Nonconverged solves and failed accuracy targets remain visible; only settings satisfying the stated numerical checks enter a frontier.

Baselines. The main linear-PDE comparisons use CG, including the L-shaped domain. Burgers uses Newton–BiCGStab and Navier–Stokes uses the recorded CNAB2 time integrator; these nonlinear evolution problems are not relabelled as CG solves. The main result tables compare NM-ROM only with these full-order solvers. Full solver configurations, checkpoint identities and numerical audits are retained with the source evidence. The appendix contains method, configuration and validation details needed to interpret the main comparisons.

6 NUMERICAL EXPERIMENTS AND RESULTS

We first compare accuracy and runtime across problems, then examine settings from a single trained model and the controls that produce them. Errors are percentages and costs are median query times. Speedup $S = T_{\text{FOM}}/T_{\text{method}}$ is always against the named comparator measured in the same allocation. Tables identify final and development cohorts; a development result is not a final accuracy claim.

6.1 COMPARISON AGAINST FULL-ORDER SOLVERS

Two-dimensional problems. Table 1 compares NM-ROM with CG on Poisson, heat and reflective waves. Increasing resolution can improve the speed ratio because the reduced computation grows more slowly than the full-grid iteration. Correction improves Poisson and wave displacement accuracy, but its runtime cost differs substantially between the two problems. The measured CG solutions remain more accurate than the corresponding NM-ROM settings. Wave displacement gains do not imply a passing velocity or energy error.

On Burgers, Table 2 compares NM-ROM with Newton–BiCGStab. At 256^2 , correction improves same-grid error, but NM-ROM does not beat the displayed FOM in accuracy or runtime. The earlier

Table 1: Two-dimensional NM-ROM versus CG: collected development results. N is intervals per axis. Error is worst relative L^2 (%); time is median GPU query time (ms). Each row uses the fastest tested passing CG with no greater displayed error. Poisson and heat use relative tolerance 10^{-2} ; wave time steps and tolerances are listed in Appendix C.1. Heat uses an earlier audited checkpoint; wave NM-ROM fails its full-state target.

Problem	N	Method	Error (%)	GPU ms	CG error (%)	CG ms	S
Poisson	256	NM-ROM, $q = 0$	3.1567	6.7031	0.1527	22.2898	$3.33\times$
Poisson	256	NM-ROM, $q = 256$	0.9689	7.1213	0.1527	22.2898	$3.13\times$
Poisson	1024	NM-ROM, $q = 0$	3.1495	3.9406	0.0722	60.0170	$15.23\times$
Poisson	1024	NM-ROM, $q = 256$	0.9648	4.3188	0.0722	60.0170	$13.90\times$
Wave	256	NM-ROM, $q = 0$	6.1035	183.5155	0.4008	142.5111	$0.78\times$
Wave	256	NM-ROM, $q = 32$	2.8017	2520.1766	0.4008	142.5111	$0.06\times$
Wave	1024	NM-ROM, $q = 0$	6.1066	192.4624	5.1413	583.0259	$3.03\times$
Wave	1024	NM-ROM, $q = 32$	2.8094	2529.4450	0.4010	1043.3681	$0.41\times$
Heat	256	NM-ROM	4.5557	11.3414	0.9255	6.9389	$0.61\times$
Heat	1024	NM-ROM	4.5555	12.2062	0.7707	59.1780	$4.85\times$

1024² paired study below also reports speed against a tight Newton control; relaxing that control substantially reduces the gain. These development solves permit stalls and do not establish stationarity. The full tested high-resolution panels remain in the source records.

Table 2: Burgers2D development comparisons. Upper panel: 256². Worst evolved and all-times errors use the initial-field norm. Upper-panel speedups use the displayed Newton–BiCGStab control: nonlinear tolerance 10^{-3} , $\Delta t = 0.005$. Faster FOM settings remain in the full panel. The corrected EQ rule is single-draw; the uncorrected rule has repeated construction confirmation.

Method	Evolved error (%)	All-times error (%)	GPU ms	S
Newton–Krylov	0.0489	0.0489	31.788	$1.00\times$
NM-ROM $q = 0$	1.8891	2.5629	40.359	$0.79\times$
NM-ROM $q = 256$	0.5129	0.9053	746.020	$0.04\times$

Earlier 1024 ² development panel (separate allocation):			
Method	Error (%)	GPU ms	FOM/NM-ROM
NM-ROM	3.908	41.677	—
Tight Newton–BiCGStab	2.142	605.747	$14.53\times$
Relaxed Newton–BiCGStab	2.390	68.044	$1.63\times$

Lower panel: four cases, three repeats, job 3534502; initial-normalised worst error against a refined reference. Both FOMs use $\Delta t = 0.005$ and FFT preconditioning. Tight nonlinear/linear tolerances are $10^{-6}/10^{-8}$; relaxed are $10^{-2}/0.5$. Both methods meet the 5% accuracy target, with unequal measured errors; NM-ROM permits stalled exits. FOM/NM-ROM divides each FOM time by the NM-ROM time in this lower panel.

Three-dimensional problems. Tables 3 and 4 put the same comparisons in the main results. Each column block uses one allocation and one stated reference convention. The Poisson CG control uses the efficient solver without timed audit-history recording. Heat’s accepted error/runtime panel is shown, with CG speedups left unreported until its paired CG measurements are incorporated. Evaluation scope is stated in Appendix C.2.

Table 3: Three-dimensional linear PDEs, development: Poisson (left) and heat (right), 32 intervals per axis. Errors are worst same-grid relative L^2 percentages; heat uses current-normalised evolved error. Corrected ranks are $q = 96$ for both, with latent dimensions 16 and 32 respectively. Poisson speedup is against CG at relative tolerance 10^{-2} . Heat CG is pending (—).

Method	Poisson3D (development)			Heat3D (development)		
	Error (%)	GPU ms	S	Error (%)	GPU ms	S
NM-ROM, uncorrected	0.589	2.450	$1.06\times$	6.013	34.771	—
NM-ROM, corrected	0.161	2.632	$0.986\times$	1.379	114.343	—
FOM	0.140	2.594	$1\times$	—	—	—

Table 4: Three-dimensional nonlinear PDEs: Burgers final (left, 33 nodes per axis) and Navier–Stokes development (right, 32 periodic points per axis). Errors are worst evolved same-grid relative L^2 percentages, normalised by the initial field. Corrected ranks are $q = 192$ and 256 . Burgers uses Newton–BiCGStab with $\Delta t = 0.01$, nonlinear tolerance 10^{-2} and inner tolerance 0.5 ; NS uses CNAB2 with $\Delta t = 0.01$. Every S uses that block’s displayed FOM.

Method	Burgers3D (final)			NS3D (development)		
	Error (%)	GPU ms	S	Error (%)	GPU ms	S
NM-ROM, uncorrected	16.263	219.161	$0.038 \times$	13.535	1253.205	$0.0025 \times$
NM-ROM, corrected	4.399	364.677	$0.0229 \times$	12.534	4622.722	$0.000677 \times$
FOM	2.312	8.337	$1 \times$	0.628	3.130	$1 \times$

Burgers3D is the frozen primary-seed result on reserved cases. Its correction ladder reduces error on every case for both evaluated initializations, but the displayed FOM remains faster and more accurate. The refined-reference checks fail the prospective physical-error budgets, so this result establishes same-grid accuracy only. The second initialization and all timing repetitions remain in the source records described in Appendix C.2. Navier–Stokes shows a smaller correction benefit and misses its accuracy target. Its displayed data are development measurements, pending final confirmation.

6.2 BEST CONFIGURATIONS PER PROBLEM AND RESOLUTION

The uncorrected and corrected rows use the same trained model within each problem. They represent different deployment settings, not separately trained fast and accurate networks. Architecture, correction ordering and numerical protocol are fixed before final testing. Table 5 isolates correction rank on Burgers2D with the residual test space held constant; increasing the rank spends more runtime to reduce error. A setting is called a passing fast setting only if it meets the declared numerical and accuracy requirements.

Table 5: Burgers2D correction settings from one frozen model at 256^2 . The test count is fixed at $M = 1088$; residual evaluation is dense. Errors are worst evolved same-grid percentages on development cases, and times are paired median GPU query times. All rows meet the numerical stopping rule.

Correction rank q	Relative L^2 error (%)	GPU query time (ms)
0	1.2657	848.0
64	1.0593	1359.4
128	0.8711	1886.5
256	0.5194	4377.9

L-shaped Poisson provides a separate geometry comparison, using CG throughout Table 6. At the finer mesh, the NM-ROM is faster at larger error than CG. Complete-query timing here includes input and output transfer, so it is not mixed with the resident-device timings above.

Table 6: L-shaped Poisson versus CG, development. Error is worst same-grid relative L^2 (%). Complete-query time includes host transfers. Each mesh uses its displayed CG tolerance and runtime as a common speedup denominator.

Mesh	Method	Error (%)	Complete ms	S
256^2	CG 0.01	0.637	11.644	$1.00 \times$
256^2	NM-ROM $q = 0$	3.861	2.879	$4.04 \times$
256^2	NM-ROM $q = 64$	2.131	3.028	$3.84 \times$
512^2	CG 0.01	0.385	29.127	$1.00 \times$
512^2	NM-ROM $q = 0$	3.852	4.716	$6.18 \times$
512^2	NM-ROM $q = 64$	2.123	4.801	$6.07 \times$

6.3 WHICH KNOB TO TURN

Correction rank changes representation capacity; tolerance and iteration limits control numerical work. EQ sample count controls residual-evaluation cost; Table 7 compares it with dense evaluation. The fixed-test-space Burgers2D ladder spans $2.44\times$ in error for $5.16\times$ in runtime. A smaller test space gives only $1.22\times$ error improvement, which is why test count is not silently varied with rank.

Table 7: Tabular comparison of dense and empirical-quadrature (EQ) residual evaluation on Burgers2D. Error is worst evolved relative L^2 (%); time is median GPU milliseconds from one allocation. The last column is dense time divided by EQ time. Dashes mean that no paired dense measurement was collected. Rules at $q = 0, 16, 32$ have repeated-construction confirmation; higher-rank constructions are marginal (Table D.2).

Correction rank q	Dense error (%)	Dense ms	EQ error (%)	EQ ms	Dense / EQ
0	1.8890	292.87	1.8891	59.07	$4.96\times$
16	—	—	1.4270	80.61	—
32	—	—	1.2493	97.70	—
64	1.0843	624.93	1.2275	120.63	$5.18\times$
128	0.8930	1190.49	0.8936	246.87	$4.82\times$
256	0.5194	3915.14	0.5389	722.21	$5.42\times$

The separate scheduled Burgers2D ladder was tested after freezing on a held-out cohort: all 4 checkpoints give monotone error reduction, with top-rank errors of 0.59–0.68 %; 3 meet the full pre-registered criterion. The incumbent’s uncorrected solve reaches 10.1120 % on a difficult held-out case, which remains in the reported maximum. Table D.1 retains every checkpoint.

Representation and numerical cost. EQ is distinct from correction rank: it approximates residual evaluation and is used only where validated. Repeated construction confirms the Burgers2D rules at $q = 0, 16, 32$; higher-rank rules retain their single-draw or marginal qualification. The current 3D results are dense and do not establish EQ acceleration. Tables 2, 5 and 7 separate method comparison, correction settings and quadrature cost; construction validation is retained in the appendix.

Limitations. Some comparisons are developmental and cohorts are small. The fixed-test-space Burgers2D study has one checkpoint; its scheduled study and Burgers3D have separate seed checks. Nonlinear initialization can fail, and same-grid improvement does not remove discretisation error. Wave full-state accuracy, NS accuracy and Burgers3D physical refinement remain limitations. These results do not establish a universal speedup or convergence guarantee.

7 CONCLUSION AND FUTURE WORK

We presented a tunable NM-ROM whose nested corrections change approximation capacity after training. The learned bank and nonlinear head support a compact weak-form solve, while precomputation and validated quadrature control its evaluation cost. The experiments separate correction-rank accuracy gains from numerical stopping effects and measure runtime against explicitly named baselines. Burgers supplies the clearest tuning evidence; the linear and nonlinear FOM comparisons delimit the current method’s accuracy and cost. Future work concerns more reliable nonlinear initialization, resolved advection-dominated tests and extension beyond structured meshes.

REPRODUCIBILITY STATEMENT

Tables and numerical prose are generated from retained run records by the accompanying scripts. Source hashes, checkpoints, solver settings, cohorts, timing repetitions and audit results identify each comparison (Appendix C). The source archive retains failed settings; the paper distinguishes development selection from held-out evaluation.

AI USE STATEMENT

Language-model assistants supported method and experiment development, implementation, result analysis, auditing, and manuscript drafting and editing. Numerical results were computed by the recorded solver runs; table generators reproduce the reported values from those records. The authors are responsible for verifying the methods, results and text.

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A METHOD DETAILS

The formulas below give the scalar two-dimensional Dirichlet instances. Per-study solver constants apply to these recorded implementations; three-dimensional and periodic-vector scope is stated in Appendix C.2.

A.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0C_q = QR$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0h_\theta(z))$ and the minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp Dh_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of B_0C_q .

A.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa Au$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag}\left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i}\right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . Two points are structural. The right-hand side is built from the *decoded current state* at every step: carrying the quantity the previous step already made small freezes the recursion and reproduces a one-step solution to round-off. And z_{n+1} enters through h_θ , so the step is a nonlinear least-squares problem, not a linear solve. At $q = R$ the step is linear in the coefficients and the whole trajectory is the exact modal propagation of the bank coefficients, with no head and no iteration; this is the “top rung” of the heat and wave ladders. The deployed path factors the damped normal matrix by Cholesky; a variant assembles the Gram matrix $S = B_0^\top B_0$ offline and forms $H = Dh_\theta^\top S Dh_\theta$ directly, which is exact and is reported as an algebraic ablation of the same step.

A.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (8)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t(N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$

gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (9)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the retained nodes, or the precomputed quadratic form $\frac{1}{2} c^\top Q c$ with Q the symmetrised tensor $T_{iab} = \sum_x P_{ix} G_{xa} (DG)_{xb}$, which is an identity only where $u > 0$ at every stencil node of every trial state the solver visits. The last is an algebraic route we verified in-job (residual and Jacobian parity below 10^{-11} relative on sign-satisfying states) and do not time in this paper. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller.

A.4 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (10)$$

over the held-out reachable states described below; the primary bar is the held-out ρ of the incumbent $q=0$ rule at the state carrying the first-interval penalty, measured in the q-diag cell and adopted as the primary bar in the q-ridge design before any certification job ran; the tight bar was declared before b-eqtop ran. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit} stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (11)$$

rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification cell the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (10) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

A.5 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\max})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per cell and are reported per cell. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. The completion rule that accepts an attained initial fit (§3.2) was pre-registered in the panel cell’s design before the job ran; complete exit flags remain in the source records.

B ARCHITECTURE AND ONLINE DATA FLOW

C EXPERIMENTAL CONFIGURATION AND REPRODUCIBILITY

The problem families and configurations below identify the two-dimensional studies. Three-dimensional meshes, correction ranks, reference conventions and FOM settings are stated beside

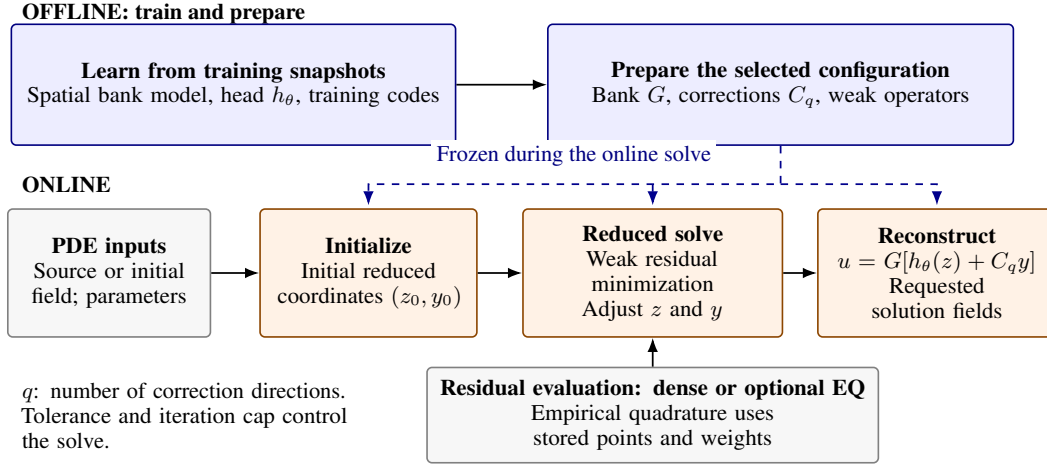


Figure B.1: NM-ROM from training to prediction. Blue components are prepared before the query and remain frozen; orange boxes compute the online solution. The inputs initialize reduced coordinates, which are adjusted to minimize the weak PDE residual before reconstructing the requested fields. Initialization and the reduced solver are PDE-specific; linear-PDE corrections can be eliminated analytically. Time-dependent problems repeat the reduced step, with reconstruction at requested output times. Empirical quadrature (EQ) is an optional residual evaluation, used in the Burgers2D EQ panels; the current 3D and wave panels do not use it. Correction rank changes the representation, whereas EQ changes residual evaluation. Baselines are evaluated independently and are not stages of this pipeline.

their main results. Within each study, training precedes evaluation and the selected bank, head and correction directions remain frozen. Deployment changes reduced coordinates, not network weights. Recorded training configurations and available offline costs are retained with the source evidence; unrecorded training costs are not inferred.

Table C.1: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded. The Burgers sealed cohort has been opened and is reported in Table D.1; other rows describe their recorded development and validation cohorts.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 (ladder 64–1024)	$\Delta t = 0.005$, backward Euler, sign- upwind	$K = 16$, $R = 512$	refined 4096 ² , $\Delta t = 0.00015625$	6 development cases; 32 held-out (tuning); sealed cohort opened once (job 3804465)
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	$256^2, 1024^2$	none (elliptic)	$K = 16$, $R = 128$ (incumbent); $K = 32$, $R = 512$ $k = 8$, $R = 32$	exact discrete (DST); 2048 ² refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$K = 32$, $R = 64$	exact modal	12 development cases (ear- lier cell, job 3511417)
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	$64^2, 256^2$, 1024^2	RK4 on the manifold; exact modal propaga- tion for the bank	none	direct DST	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	$256^2, 512^2$	none	$K \in \{16, 32\}$, $R \in \{256, 512, 514\}$	sparse direct (Su- perLU)	3072 / 256 / 32 sources (train / selection / develop- ment)

C.1 CG SETTINGS AND TIMING

The main two-dimensional rows use the fastest retained passing CG setting whose measured error does not exceed the NM-ROM error. This is a selection from a finite tested set. Poisson and heat use relative tolerance 10^{-2} . At the finest wave mesh, the uncorrected comparison uses $\Delta t = 0.005$ and tolerance 10^{-2} ; the corrected comparison uses $\Delta t = 0.0025$ and tolerance 10^{-6} . Heat uses an earlier audited checkpoint. Wave displacement alone fails to establish the full-state target.

Table C.2: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim$ $\log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/ mr-burgers2d/ engines.py: params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim$ $\log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage- precision/ms_ parametric.py: sample_params
Poisson, L-shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	as above; sources centred in the removed quadrant rejected	experiments/ lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (single_bc_poly_gaussian_v1); Crank-Nicolson $\Delta t = 0.025$	heat linear-bank report, job 3511417
Wave 2D (reflective)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_i \sim U(0.36, 0.42)$, centre $c_i \sim$ $U(s_i + 0.025, 1 - s_i - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_i \sim U(0.12, 0.16)$, advective velocity $v_i \sim$ $U(-0.5, 0.5)$ (zero every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/ multiresolution- wave/audit_ dynamics.py: parameter_rows

Every reported cost and error come from the same invocation. Timings use GPU warm-up and synchronization, double precision, and the highest matrix multiplication precision. All repetitions, including outliers, remain in the median; no time is borrowed from a different allocation. Requested full fields and initialization are charged. Host transfer is included only where the main caption labels complete-query cost. Tight and relaxed Burgers controls are separate named baselines, not interchangeable accuracy matches.

C.2 THREE-DIMENSIONAL EVALUATION SCOPE

Burgers uses its frozen primary seed on the reserved final cohort. The main Poisson, heat and Navier-Stokes rows retain their explicitly labelled development snapshots; later evaluations require separate integration. The Burgers final refinement check fails its physical-error budget, so its table establishes same-grid reduction error only. Navier-Stokes misses its accuracy target. Neither result is evidence of a universal speedup. Heat CG remains absent from the displayed snapshot; no ratio is inferred.

Scalar three-dimensional Poisson, heat and Burgers extend their spatial operators across all three axes. Navier-Stokes uses periodic vector fields, incompressibility projection and the recorded CNAB2 time integrator; its FOM is not a CG solve. The reflective-wave NM-ROM uses its recorded latent time integration, while its iterative FOM comparator uses implicit midpoint. The scalar Dirichlet formulas in Appendix A do not replace these PDE-specific implementations.

C.3 SOURCE RECORDS

The accompanying source package retains the full experiment archive, including failed configurations, retractions, training records, checkpoints, case-level fields, timing repetitions and independent audits. The PDF contains the evidence needed for its stated claims rather than every experiment in that archive. Table generators and frozen evidence identify the displayed values through these machine-readable manifests:

- tables/provenance.json: original two-dimensional studies.
- tables/rewrite-provenance.json: paired CG comparisons.
- tables/burgers-iterative-provenance.json: the earlier fine-grid Burgers tight and relaxed controls.
- tables/main-experiments-provenance.json: the displayed three-dimensional snapshots and named FOM denominators.

Each manifest records source hashes; its corresponding evidence retains solver configuration, checkpoint identity and allocation metadata. Development selection and final evaluation are distinct. Solver exit flags document numerical stopping, not a proof of global optimality.

D VALIDATION OF THE CORRECTION AND QUADRATURE STUDIES

Table D.1: The sealed cohort (lane b-seeds, job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(K + q)$ ladder, the incumbent’s sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the lane’s sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The incumbent’s $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the incumbent alone (5.35) and the convergence criterion fails on *seed2* at $q = 64$ (3 budget exits).

q	M	incumbent sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 ± 0.6357	2.0002 ± 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 ± 0.1297	1.5988 ± 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 ± 0.1422	1.3727 ± 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 ± 0.1099	1.2130 ± 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 ± 0.0619	0.9562 ± 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 ± 0.0345	0.4987 ± 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
incumbent	3804465	yes	yes	yes	$14.89 \times$	$17.93 \times$	yes
seed1	3804465	yes	yes	yes	$3.52 \times$	$13.25 \times$	yes
seed2	3804465	yes	yes	no	$4.81 \times$	$13.17 \times$	no
seed3	3804465	yes	yes	yes	$5.22 \times$	$14.62 \times$	yes

Table D.2: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows read from the lane summary.json (final).

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table D.3: How the corrections are solved on Burgers, one job (job 3734098, NVIDIA A100 80GB PCIe): joint LM on (z, y) , block-damped, and plain variable projection at $q = 64$ and 128. Same error where all converge; the block-damped step is the one kept.

arm	M	worst all-times %	budget exits	all stationary	device ms
$q=64$, joint LM on (z, y)	320	2.1489	6	no	1119.7
$q=64$, block-damped	320	2.1489	0	yes	619.2
$q=64$, plain variable projection	320	2.1489	0	yes	9527.9
$q=128$, plain variable projection	256	1.8116	297	no	47566.9
$q=128$, block-damped	256	1.8116	0	yes	825.5

Reading the validation tables. Correction rank q is the number of added coefficient directions; M is the number of weak test modes and m the quadrature node count. A scheduled ladder changes M with q ; the main fixed-test-space study holds M constant. A sealed cohort is opened only after choices are frozen. Confirmed quadrature passes every independent reconstruction of the rule; marginal means some reconstructions fail. These statuses concern rule construction, not just the error of one selected trajectory. Same-grid error and error against a refined reference measure different quantities.